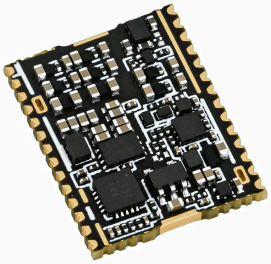
Stamp Timer Power2

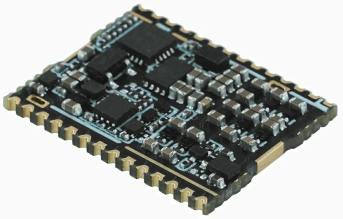
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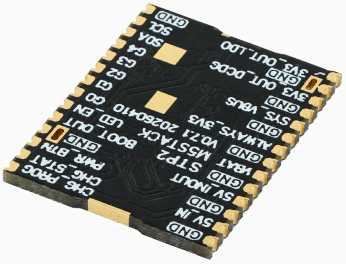


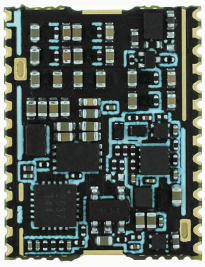


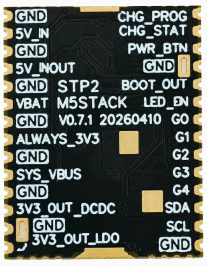
















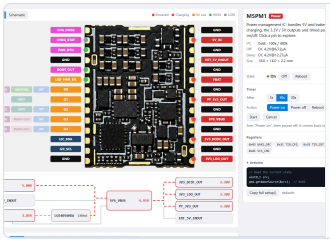




Description

Stamp Timer Power2 is a power management module for low-power embedded devices. It integrates the M5PM1 power management IC and adopts a multi-stage power control architecture that enables power for the main controller and various peripherals on demand, effectively reducing overall standby power consumption and extending battery life. The module supports DC 5V input/output control, DC 3.7V lithium battery power supply, and input voltage detection. It also integrates a lithium battery charging circuit with two adjustable charging current levels. Featuring a 1.27mm castellated-hole package, the module provides **5 programmable IOs** in addition to the basic power input/output control pins. These IOs can be configured as GPIO, ADC, PWM, RGB LED, IRQ, and other functions. The module also supports timed wake-up and external input interrupt wake-up, making it suitable for low-power systems with on-demand power-up, periodic operation, and event-triggered operation. It is well suited for embedded device power management, smart homes, and low-power IoT devices.

Tutorial



Stamp Timer Power2 Online Interactive Demo

This page provides an interactive online demonstration of the pins, power paths, and registers of Stamp Timer Power2 (M5PM1), with interactive debugging and example code.

Features

- Built-in M5PM1 power management IC
- Built-in lithium battery charging IC with two adjustable charging current levels (650mA/180mA), suitable for different batteries
- 5x programmable expansion IO (GPIO / ADC / PWM / RGB LED / IRQ)
- Supports DC 5V / DC 3.7V battery input power
- I2C communication interface (0x6E)
- 1.27mm-28Pin castellated-hole package
- Peripherals & Control
  - Power button input signal interface
  - Power indicator signal interface
  - Boot control interface for MCU download mode control

Includes

- 1 x Stamp Timer Power2

Applications

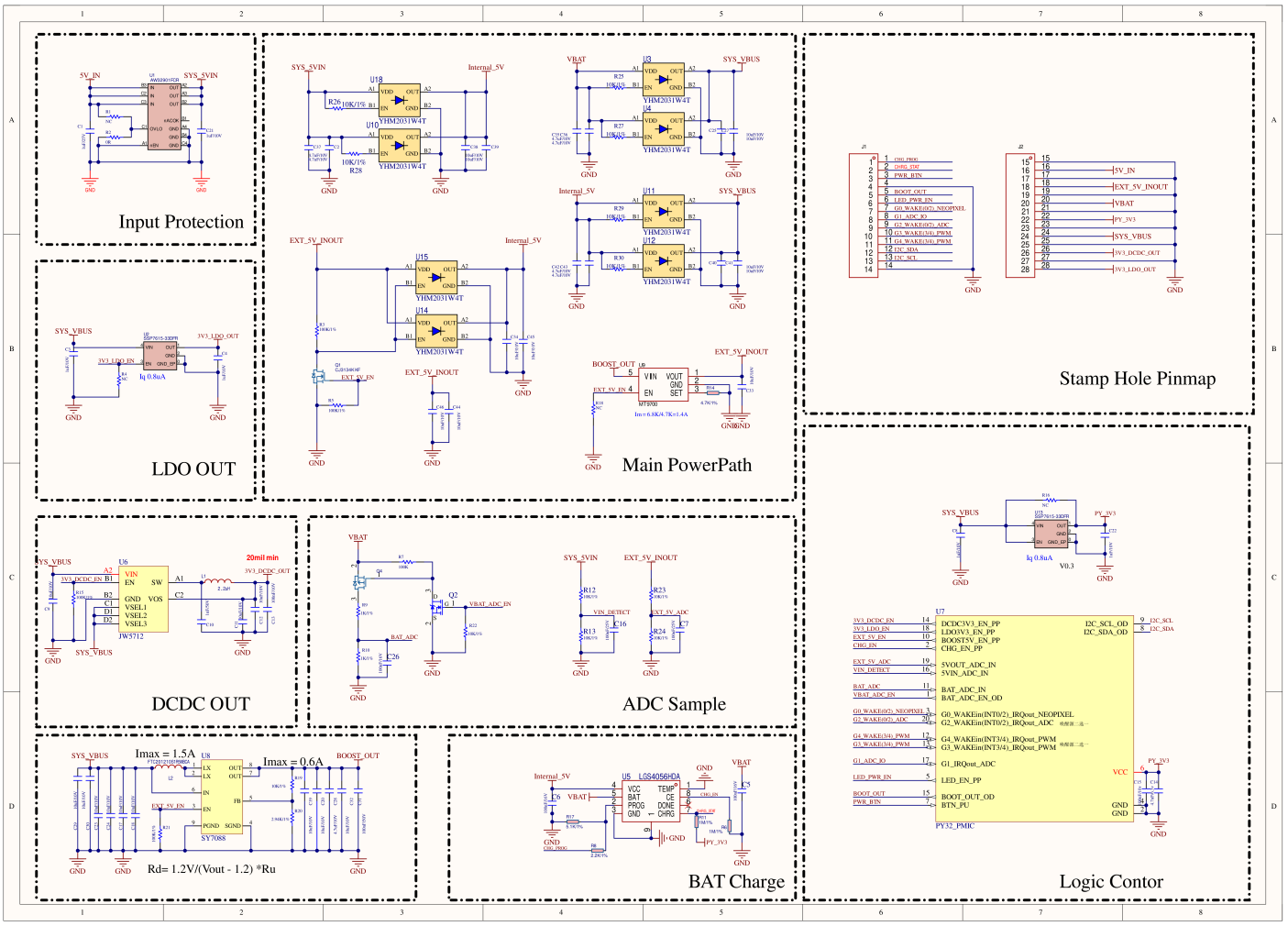
- Embedded device power management
- Smart home
- Low-power IoT devices

Specifications

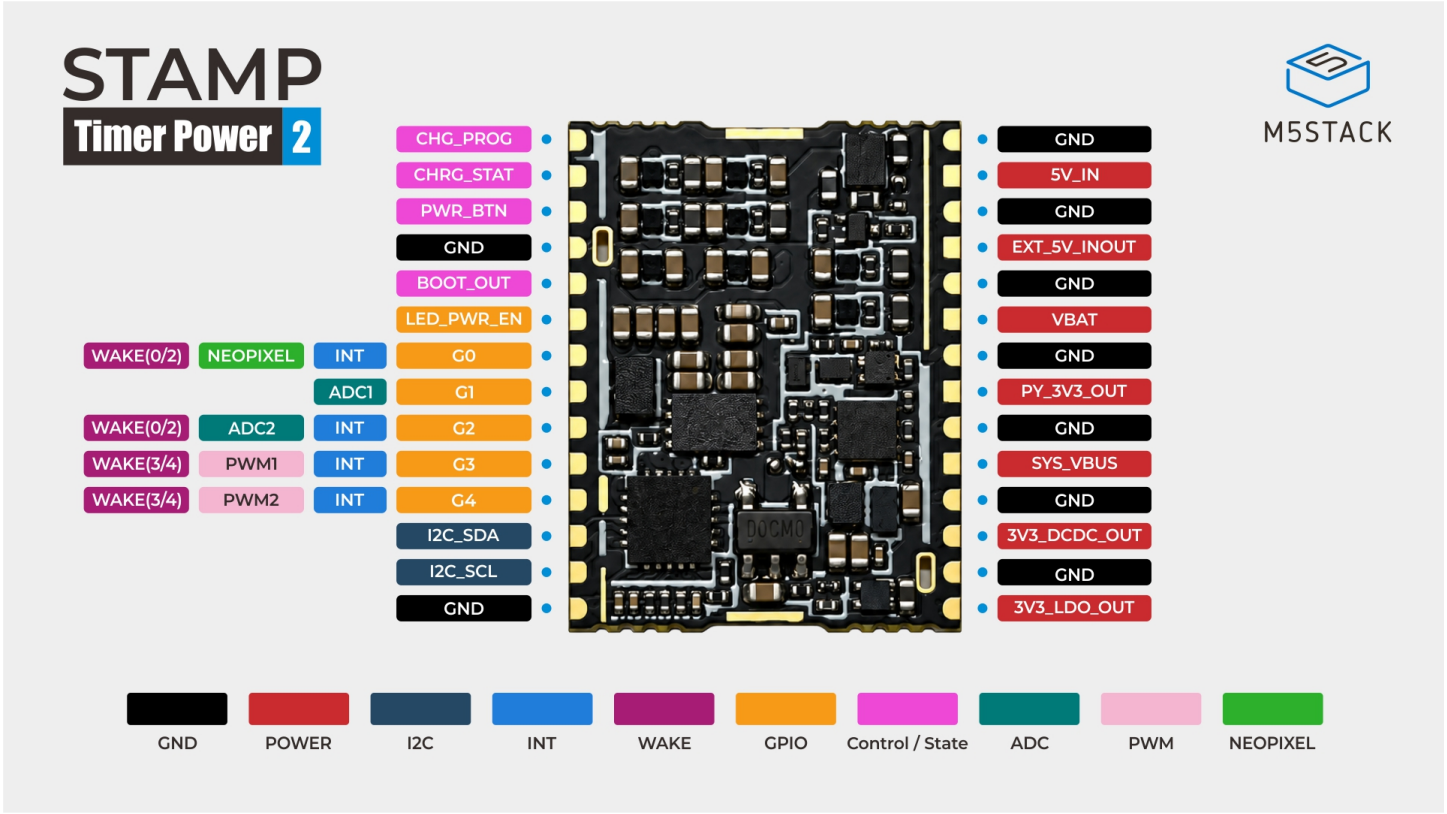
Specification	Parameter
Power Management IC	M5PM1
Communication Interface	I2C @ 0x6E, supports 100 kHz (default) / 400 kHz
Power Consumption	Shutdown: DC 4.2V@6.72uA Low-power sleep: DC 4.2V@12.27uA
Battery Charging IC	LGS4056HDA, provides two adjustable charging current levels: DC 3.3V@650mA / DC 3.3V@180mA
Load Capacity	3V3_DCDC_OUT: DC 3.3V@0.6A 3V3_LDO_OUT: DC 3.3V@0.4A PY_3V3: DC 3.3V@0.4A EXT_5V_INOUT: DC 5V@1A
Product Size	18.0 x 14.0 x 2.2mm
Product Weight	0.7g
Package Size	138.0 x 93.0 x 5.0mm
Gross Weight	3.8g

Schematics

- Stamp Timer Power2 Schematics PDF

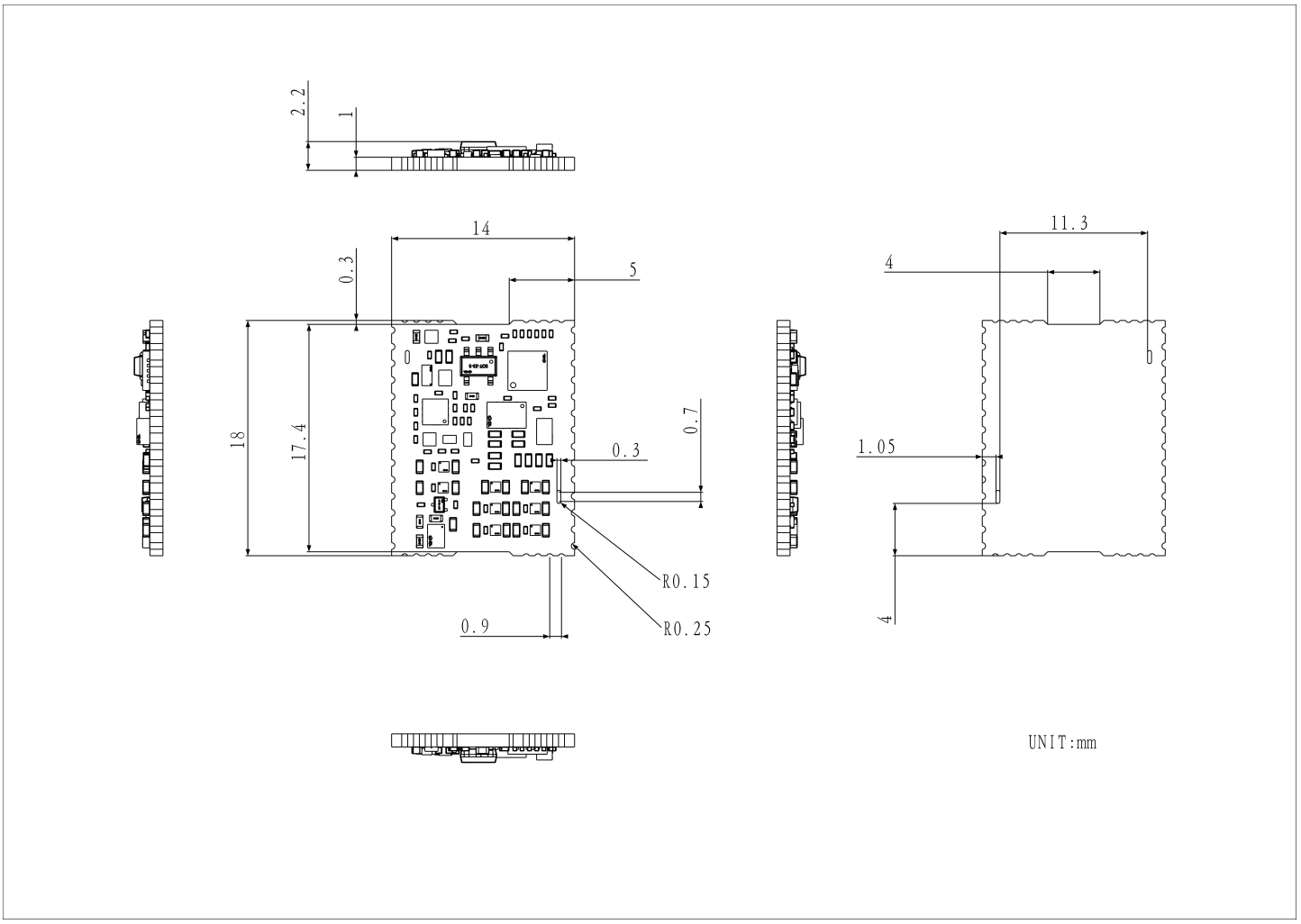

Input Protection
LDO OUT
Main PowerPath
Stamp Hole Pinmap
DCDC OUT
ADC Sample
BAT Charge
Logic Contor
Rd= 1.2V/(Vout - 1.2) *Ru

PinMap


STAMP
Timer Power 2
M5STACK
CHG_PROG
CHRG_STAT
PWR_BTN
GND
BOOT_OUT
LED_PWR_EN
WAKE(0/2) NEOPIXEL INT G0
ADC1 G1
WAKE(0/2) ADC2 INT G2
WAKE(3/4) PWM1 INT G3
WAKE(3/4) PWM2 INT G4
I2C_SDA
I2C_SCL
GND
GND
5V_IN
GND
EXT_5V_INOUT
GND
VBAT
GND
PY_3V3_OUT
GND
SYS_VBUS
GND
3V3_DCDC_OUT
GND
3V3_LDO_OUT
GND
POWER
I2C
INT
WAKE
GPIO
Control / State
ADC
PWM
NEOPIXEL

Model Size

- Stamp Timer Power2 Model Size PDF



PCB

- Stamp Timer Power2 PcbDoc
- Stamp Timer Power2 KiCad Footprint Library

Structure

- Stamp Timer Power2 KiCad 3D

Datasheets

- LGS4056HDA

Softwares

Arduino

- M5PM1 Arduino Driver Library

UiFlow2

- Stamp Timer Power2 UiFlow2 Documentation

Communication Protocol

- M5PM1 Power Management IC

View Protocol

Others

- Stamp Timer Power2 Online Interactive Demo

Video